HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
Faculty for High Quality Training
Program name: Group of Mathematics

FINAL EXAMINATION SEMESTER 2
ACADEMIC YEAR 2022-2023
Course name: CALCULUS 2
Course code: MATH132501E
Number of pages: 02 pages.
Duration: 90 minutes. Date of exam: 23/05/2023
Open book.

Question 1 (total mark **1/10)**

Find the area of the region enclosed by the polar - curve

$$r = 5 - 3\cos\theta.$$

Question 2 (total mark **1/10)**

Find the value of the following improper integral, if it converges:

$$\int_2^{\infty} \frac{t}{(t^2-1)^3}\,dt.$$

Question 3 (total mark **1/10)**

Let D be the region bounded by $y = 4x - x^2$ and the x-axis. Find the volume of the solid generated by revolving D about the line $x = -1$.

Question 4 (total mark **1/10)**

Test the series for convergence

$$\sum_{k=1}^{\infty} \left(\frac{k-2}{k}\right)^k .$$

Question 5 (total mark **1/10)**

Find the value of the geometric series

$$\sum_{k=0}^{\infty} \frac{2023^k}{10^{4k+1}}.$$

Question 6 (total mark **1.5/10)**

Find the convergence set for the power series

$$\sum_{k=1}^{\infty} \frac{k^2+20k}{23^k}(x+3)^k.$$

Question 7 (total mark **1/10)**

Find the coefficient of x^4 in the Maclaurin series of the function

$$f(x) = x\sin(2x).$$

Question 8 (total mark **1.5/10)**

Find the particular solution of the differential equation

$$x\frac{dy}{dx} + 2xy - xe^{-x} = 0, \qquad x > 0,$$

which satisfies the initial condition $y(1) = 1/e$.

Question 9 (total mark **1/10)**

In the three-dimensional space, find the area of the triangle OAB where $A(1,2,1)$, $B(-3,1,0)$ and $O(0,0,0)$.

Note: Proctors are not allowed to give any unauthorized explanation.

Expected Learning Outcomes		**Questions**
CLO1	Explain the concept of convergence of improper integrals and of series of numbers, and convergence set of power series.	2,4,5,6,7
CLO2	Use methods of integration. Test an improper integral or a seriesfor convergence, and find the convergence set for a power series.	1,3,4,5,6,7
CLO3	Construct mathematical models using the first-order linear differential equations.	8
CLO4	Evaluate the dot product and the cross product of two vectors in $\mathbb{R}^3$.	9
CLO5	Establish formulas for area of a region in plane, volume and arclength by using definite integrals.	1,3

15/05/2023

Approved by program chair

(signed and named)

TS. Phạm Văn Hiển